

Supplementary Material

This is the supplementary material for the paper “Provable Fairness Repair for Deep Neural Networks”. It includes proofs of the theorems presented in the main text, details of the fairness specifications used in our experiments, and provides the full results of some experiment that could not be included in the main paper due to space limitations.

PROOFS OF THEOREMS

We provide the proofs for the theorems stated in the paper.

Theorem 1. Let $\mathcal{D}_r = \{\mathbf{x}^i\}_{i=1}^n$ be a set of inputs, each associated with a similarity neighborhood $\mathcal{S}(\mathbf{x}^i)$, and let $f = f_L \circ \tilde{f}_{1:L}$ be a DNN. Then any feasible solution $(\Delta\mathbf{W}, \Delta\mathbf{b})$ to the problem (14) induces a repaired final layer $\tilde{f}_L$ such that the composite model $\tilde{f} = \tilde{f}_L \circ \tilde{f}_{1:L}$ provably satisfies individual fairness on all $\mathcal{S}(\mathbf{x}^i)$, $i \in [n]$.

Proof. Consider any feasible solution $(\Delta\dot{\mathbf{W}}, \Delta\dot{\mathbf{b}})$ to problem (14). Recall that the sets $\mathcal{H}^i$ satisfy

$$\mathcal{H}^i = \{f_{1:L}(\mathbf{x}) \mid \mathbf{x} \in \mathcal{S}(\mathbf{x}^i)\} \subseteq [\underline{\mathbf{h}}^i, \bar{\mathbf{h}}^i]$$

and that the lower and upper bounds LB_i, UB_i are constructed via the extreme values of auxiliary optimization problems P_i and Q_i to satisfy

$$\begin{aligned} \text{LB}_i &\leq \min_{\mathbf{h} \in [\underline{\mathbf{h}}^i, \bar{\mathbf{h}}^i]} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} + \mathbf{b} + \Delta\dot{\mathbf{b}} \\ \text{UB}_i &\geq \max_{\mathbf{h} \in [\underline{\mathbf{h}}^i, \bar{\mathbf{h}}^i]} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} + \mathbf{b} + \Delta\dot{\mathbf{b}} \end{aligned}$$

Since $\mathcal{H}^i \subseteq [\underline{\mathbf{h}}^i, \bar{\mathbf{h}}^i]$, these bounds also hold over $\mathcal{H}^i$. By feasibility, the repair constraints ensure that either $\text{LB}_i \geq 0$ or $\text{UB}_i < 0$, which guarantees that for all $\mathbf{h} \in \mathcal{H}^i$, the repaired model output $\tilde{f}_L(\mathbf{h})$ is consistently non-negative or negative. Consequently, the composite model $\tilde{f} = \tilde{f}_L \circ \tilde{f}_{1:L}$ provably satisfies individual fairness across all neighborhoods $\mathcal{S}(\mathbf{x}^i)$, $i \in [n]$. $\square$

Theorem 2. Let $\mathcal{D}_r = \{\mathbf{x}^i\}_{i=1}^n$ be a set of inputs, each associated with a similarity neighborhood $\mathcal{S}(\mathbf{x}^i)$, and let $f = f_L \circ \tilde{f}_{1:L}$ be a DNN. Then any feasible solution $(\Delta\mathbf{W}, \Delta\mathbf{b}, \mathbf{Z}, \boldsymbol{\lambda}, \boldsymbol{\eta})$ to the problem (22) induces a repaired final layer $\tilde{f}_L$ such that the composite model $\tilde{f} = \tilde{f}_L \circ \tilde{f}_{1:L}$ provably satisfies individual fairness on all $\mathcal{S}(\mathbf{x}^i)$, $i \in [n]$. Moreover, the optimal solution of problem (22) is guaranteed to be no worse than that of problem (14) (i.e., it achieves an objective value that is no larger).

Proof. Consider $(\Delta\dot{\mathbf{W}}, \Delta\dot{\mathbf{b}}, \dot{\mathbf{Z}}, \dot{\boldsymbol{\lambda}}, \dot{\boldsymbol{\eta}})$ as a feasible solution to the MILP problem (22). We denote $\dot{\mathbf{C}} = \begin{pmatrix} \mathbf{W} + \Delta\dot{\mathbf{W}} \\ \mathbf{0}^m \end{pmatrix}$.

Recalling the dual problem constructed in Eq. (19) and the strong duality [1], we have:

$$\begin{aligned} \max_{\substack{\mathbf{A}_i^\top \boldsymbol{\lambda}_i = -\dot{\mathbf{C}} \\ \boldsymbol{\lambda}_i \geq 0}} -\boldsymbol{\lambda}_i^\top \mathbf{D}_i &= \min_{\mathbf{A}_i \mathbf{p} \leq \mathbf{D}_i} \dot{\mathbf{C}}^\top \mathbf{p} = \min_{\mathbf{h} \in \hat{\mathcal{H}}_i} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} \\ \min_{\substack{\mathbf{A}_i^\top \boldsymbol{\eta}_i = \dot{\mathbf{C}} \\ \boldsymbol{\eta}_i \geq 0}} \boldsymbol{\eta}_i^\top \mathbf{D}_i &= \max_{\mathbf{A}_i \mathbf{p} \leq \mathbf{D}_i} \dot{\mathbf{C}}^\top \mathbf{p} = \max_{\mathbf{h} \in \hat{\mathcal{H}}_i} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} \end{aligned}$$

Thus, the model outputs over $\mathcal{S}(\mathbf{x}^i)$ are bounded by $\widehat{\text{LB}}_i$ and $\widehat{\text{UB}}_i$ as:

$$\begin{aligned} \widehat{\text{LB}}_i &= \mathbf{b} + \Delta\dot{\mathbf{b}} + (-\dot{\boldsymbol{\lambda}}_i^\top \mathbf{D}_i) \\ &\leq \mathbf{b} + \Delta\dot{\mathbf{b}} + \max_{\boldsymbol{\lambda}_i} -\boldsymbol{\lambda}_i^\top \mathbf{D}_i \\ &= \mathbf{b} + \Delta\dot{\mathbf{b}} + \min_{\mathbf{h} \in \hat{\mathcal{H}}_i} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} \quad \text{s.t. } \mathbf{A}_i^\top \boldsymbol{\lambda}_i = -\mathbf{C} \\ &= \min_{\mathbf{h} \in \hat{\mathcal{H}}_i} \tilde{f}_L(\mathbf{h}) \\ &\leq \min_{\mathbf{h} \in \mathcal{H}_i} \tilde{f}_L(\mathbf{h}) = \min_{\mathbf{x} \in \mathcal{S}(\mathbf{x}^i)} \tilde{f}(\mathbf{x}) \\ \widehat{\text{UB}}_i &= \mathbf{b} + \Delta\dot{\mathbf{b}} + (\dot{\boldsymbol{\eta}}_i^\top \mathbf{D}_i) \\ &\geq \mathbf{b} + \Delta\dot{\mathbf{b}} + \min_{\boldsymbol{\eta}_i} \boldsymbol{\eta}_i^\top \mathbf{D}_i \\ &= \mathbf{b} + \Delta\dot{\mathbf{b}} + \max_{\mathbf{h} \in \hat{\mathcal{H}}_i} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} \quad \text{s.t. } \mathbf{A}_i^\top \boldsymbol{\eta}_i = \mathbf{C} \\ &= \max_{\mathbf{h} \in \hat{\mathcal{H}}_i} \tilde{f}_L(\mathbf{h}) \\ &\geq \max_{\mathbf{h} \in \mathcal{H}_i} \tilde{f}_L(\mathbf{h}) = \max_{\mathbf{x} \in \mathcal{S}(\mathbf{x}^i)} \tilde{f}(\mathbf{x}) \end{aligned}$$

where the blue inequality holds due to the soundness of the symbolic bounds, i.e., $\mathcal{H}_i \subseteq \hat{\mathcal{H}}_i$. Finally, by integrating $\widehat{\text{LB}}_i \geq -M \cdot (1 - \dot{Z}_i) \wedge \widehat{\text{UB}}_i < M \cdot \dot{Z}_i$, the model’s outputs over $\mathcal{S}(\mathbf{x}^i)$ are strictly confined to either positive or negative regimes, guaranteeing consistent classifications and thereby ensuring individual fairness.

We next prove that the optimal solution of problem (22) is guaranteed to be no worse than that of problem (14). The key point is that any feasible solution $(\Delta\dot{\mathbf{W}}, \Delta\dot{\mathbf{b}})$ of the problem (14) can be extended to a feasible solution $(\Delta\dot{\mathbf{W}}, \Delta\dot{\mathbf{b}}, \dot{\mathbf{Z}}, \dot{\boldsymbol{\lambda}}, \dot{\boldsymbol{\eta}})$ of the MILP problem (22). This is because the symbolic bounds encoded in problem (22) is tighter than the concrete bounds, which implies $\hat{\mathcal{H}}_i \subseteq [\underline{\mathbf{h}}^i, \bar{\mathbf{h}}^i]$, thus we have:

$$\begin{aligned} \min_{\mathbf{h} \in [\underline{\mathbf{h}}^i, \bar{\mathbf{h}}^i]} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} &\leq \min_{\mathbf{h} \in \hat{\mathcal{H}}_i} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} \\ \max_{\mathbf{h} \in [\underline{\mathbf{h}}^i, \bar{\mathbf{h}}^i]} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} &\geq \max_{\mathbf{h} \in \hat{\mathcal{H}}_i} (\mathbf{W} + \Delta\dot{\mathbf{W}})\mathbf{h} \end{aligned}$$

In other words, the fairness constraints in problem (22) are relaxed compared those in problem (14). Therefore, the feasible region of problem (14) is a subset of that of problem (22). Consequently, problem (22) minimizes the same objective over

a (potentially) larger feasible region, and hence its optimal value is no greater than that of problem (14). This completes the proof of the claimed optimality relationship. $\square$

- ϕ_G : This specification defines similarity between individuals whose “credit-amount” attribute lies within a fixed interval of size 100.

SPECIFICATION DESIGN DETAILS

We describe the relaxed fairness specifications used in our experiments. Two individuals are considered similar if their corresponding attributes satisfy the following conditions, regardless of differences in sensitive attributes:

- ϕ_A : Two individuals are considered similar if their “hours-per-week” attribute differs by at most 1.
- ϕ_B : Two individuals are regarded as similar when their “duration” attribute varies by no more than 1.

REFERENCES

- [1] S. P. Boyd and L. Vandenberghe, *Convex optimization*. Cambridge university press, 2004.

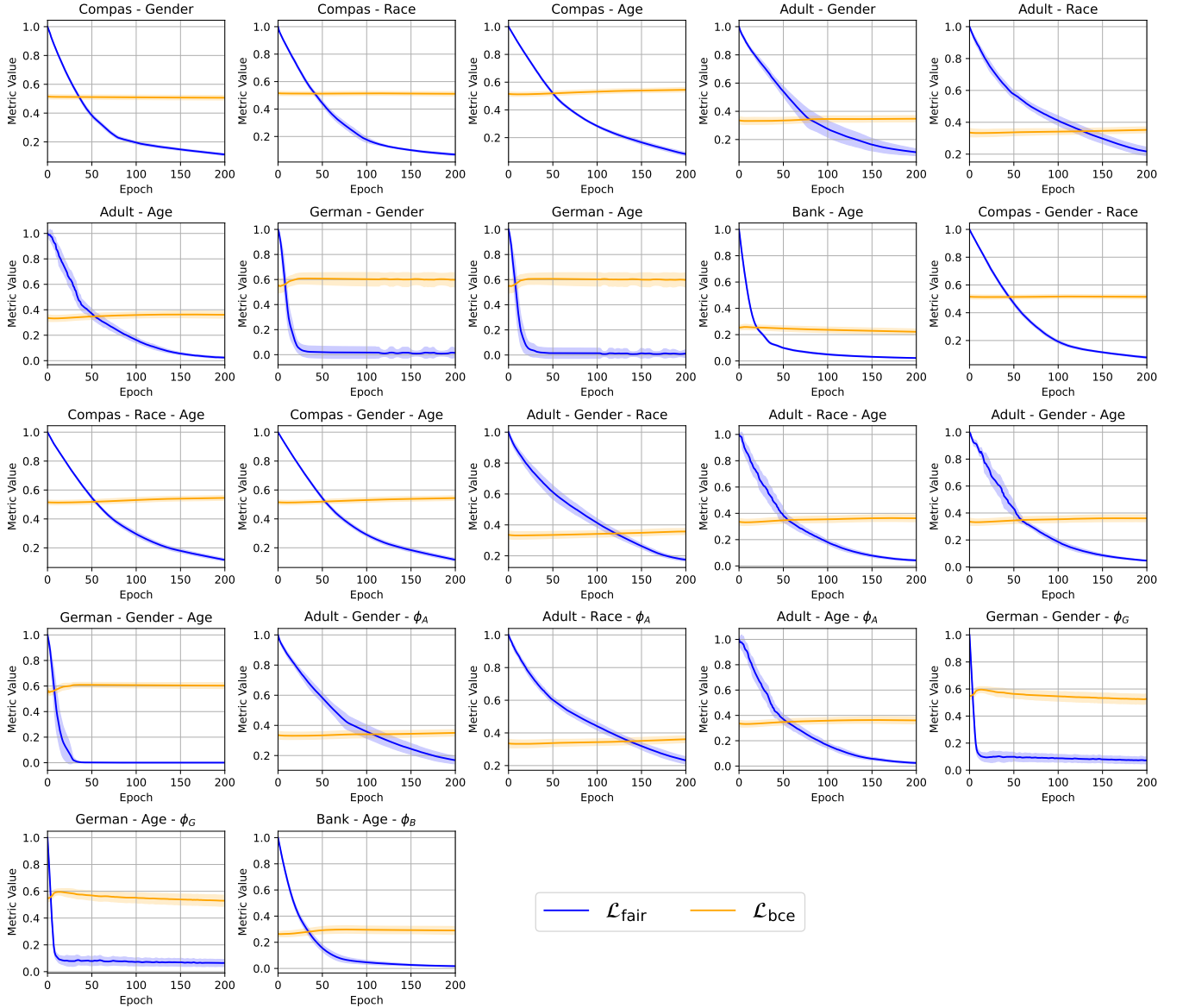


Fig. 1: Full results of loss dynamics during the progressive bounds tightening process in Step 1 of PROF, with shaded areas representing the variance across multiple runs.